LAB ASSIGNMENT 03
OBJECT ORIENTED PROGRAMMING

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Course : Object oriented programming with C++
Group : 1MCA3

1. **Create a class Student with private data members name and marks. Write a friend function to access and display these private data members.**

```
#include<iostream>
using namespace std;
class Student {
 private:
 string name;
 int marks;

 public:
 Student()
  {
 name="Mahak";
 marks= 85;
  }
 friend void display(Student s);
};
void display(Student s)
{
 cout<<"Name: "<<s.name<<endl;
 cout<<"Marks: "<<s.marks<<endl;
}
int main() {
 Student s;
 display(s);
 return 0;
}
```

```
Name: Mahak
Marks: 85

--------------------------------
Process exited after 2.143 seconds with return value 0
Press any key to continue . . .
```

2. **Create a class Number with two private data members a and b. Use a friend function to access these members and calculate and display their sum.**

```
#include<iostream>
using namespace std;
class Number {
 private:
 int a,b;
```

```
 public:
 Student()
  {
 a=10;
 b=20;
  }
 friend void calculateSum(Number n);
};
void calculateSum(Number n)
{
 int sum;
 sum= n.a + n.b;
cout<<"Sum = "<<sum;
}
int main() {
 Number n;
calculateSum(n);
 return 0;
}
```

```
Sum = 0
--------------------------------
Process exited after 0.2368 seconds with return value 0
Press any key to continue . . .
```

3. **Create a class Employee with a private data member salary. Create two objects and use a friend function to compare their salaries and display the higher salary.**

```
#include <iostream>
using namespace std;

class Employee {
private:
    float salary;

public:
Employee(float s) {
        salary = s;
    }
 friend void compareSalary(Employee e1, Employee e2);
};
void compareSalary(Employee e1, Employee e2) {
    if (e1.salary > e2.salary) {
        cout << "Higher Salary: " << e1.salary << endl;
    }
    else if (e2.salary > e1.salary)
    {
        cout << "Higher Salary: " << e2.salary << endl;
    }
    else
```

```
 {
 cout << "Both salaries are equal." << endl;
}
}
int main() {
 Employee e1(50000);
 Employee e2(60000);
 compareSalary(e1, e2);
 return 0;
}
```

```
Higher Salary: 60000

--------------------------------
Process exited after 0.2751 seconds with return value 0
Press any key to continue . . .
```

4. **Create a class Student with a static data member to keep track of the number of objects created. Create three objects and display the total number of objects.**

```
#include <iostream>
using namespace std;

    class Student {
    private:
        static int count;
    public:
         Student() {
            count++;

 }
 static void displayCount() {
     cout << "Total number of objects: " << count << endl;

    }
    };
    int Student::count = 0;
    int main() {
    Student s1;
    Student s2;
    Student                    s3;
    Student::displayCount();
    return 0;
   }
```

```
Total number of objects: 3

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Process exited after 0.2679 seconds with return value 0
Press any key to continue . . .
```

5. **Create a class BankAccount with:**

Private data members:

accountNo

balance

Static data member:

totalAccounts

```
#include <iostream>
using namespace std;
class BankAccount{
        private:
          int accountNo;
          float balance;
          static int totalAccounts;
        public:
          BankAccount(int acc, float bal)
            {
              accountNo = acc;
              balance = bal;
              totalAccounts++;
            }
        friend void displayAccount(BankAccount b);
       static void displayTotalAccounts()
            {
        cout << "Total Accounts: " << totalAccounts << endl;
             }
            };
       int BankAccount::totalAccounts = 0;
      void displayAccount(BankAccount b) {
            cout << "Account No: " << b.accountNo << endl;
            cout << "Balance: " << b.balance << endl;
            }
      int main()
            {
            BankAccount b1(101, 5000);
             BankAccount b2(102, 7000);
            BankAccount b3(103, 9000);
            displayAccount(b1);
           cout << endl;
           displayAccount(b2);
           cout << endl;
          displayAccount(b3);
          cout << endl;
          BankAccount::displayTotalAccounts();
        return 0;
            }
```

```
Account No: 101
Balance: 5000

Account No: 102
Balance: 7000

Account No: 103
Balance: 9000

Total Accounts: 3

--------------------------------
Process exited after 3.436 seconds with return value 0
Press any key to continue . . .
```